Total No of Pages 02 Roll. No.....

SUPPLEMENTARY EXAMINATION FEBRUARY-2018

HU201 ENGINEERING ECONOMICS

Time: 03: 00 Hours Max.Marks: 50

Note: *Answer any five questions. Assume suitable missing data if any.*

1(a) Define consumer demand and discuss the factors upon which demand depends. 5

1(b) If a company earned a profit of Rs.50000 during the year 1994-95. If the marginal cost and selling price of a product are Rs.10 and Rs.15 per unit respectively,find out the margine of safety. 5

2(a) Define price elasticity of demand and discuss the relationship between average revenue (AR), marginal revenue (MR) and price elasticity. 5

2 (b) A company has to replace a machine in the production line after 11 years at the cost of Rs. 60,00,000/-. It plans to deposit an equal amount at the end of every year for the next 11 years at an interest rate of 11 per cent which is compounded annually. Find the equivalent amount that must be deposited at the end of every year for next 11 years. 5

3(a) Discuss the nature of computer soft ware market in India. As it is one of highest foreign reserve earner for India, what incentive may be given by the Government to encourage Indian producer? 5

3(b) If the demand law is given by x=20/(x+1), find elasticity of demand with respect to price at the point when p=3. 2

3(c) What amount of money saved today will yield Rs. 40,000/- in third year and Rs. 55,000/- after five year at the 12% rate of interest compounded annually. 3

4(a) What do you mean by Inflation? Discuss role of Fiscal policy in control of Inflation 5

4(b) A Company wants to deposit money to create an R&D reserve. The company will get 15,00,000 every year for next 15 years for R&D. The reserve will grow at the rate of 12 per cent annually. Find out the single payment which should be made now. 5

5(a) Environmental problems are macro level problems. Discuss the role of appropriate technology in this regard. Also suggest framework to impose taxes to control environmental hazards. 5

5(b) A firm, Prime Manufacturing is planning to expand its production operation. It has identified three machines which are technologically capable to serve the purpose. The initial outlay and annual revenues with each of the machines are given below: 5

	Initial Outlay (Rs.)	Annual revenue	Life (in years)
Machine I	Rs. 5,45,000	Rs. 2,50,000	15
Machine II	Rs. 6,14,000	Rs. 3,30,000	12
Machine III	Rs. 6,00,900	Rs. 3,50,000	10

If the rate of interest is 12%, which machine the company should opt for? Find out the result by Future Worth method.

6(a) Discuss application of Engineering & Technology to improve life at slum. 5

6(b) A company must decide whether to buy machine A or Machine B 5

	Machine A	Machine B
Initial Cost (Rs.)	3,00,000	6,00,000
Useful Life (years)	4	4
Salvage value at the end of machine life (Rs.)	2,00,000	3,00,000
Annual Maintenance (Rs.)	30,000	0

At 12% interest rate, which Machine should be purchased? Use Present Worth Method.

7(a) Discuss Business Risk which you will consider while deciding location for your foreign venture. 5

7(b) A man owns a corner plot. He must decide which of the several alternatives to select in trying to obtain a desirable return on his investment. After much study and calculations, he decides that the two nest alternatives are as follows: 5

	Build Gas station	Build Soft ice-cream stand
Initial Cost (Rs.)	20,00,000	36,00,000
Annual Property Taxes (Rs.)	89,000	1,50,000
Annual Income (Rs.)	8,00,000	9,80,000
Life of Building (years)	20	20

There is no salvage value. Evaluate the alternatives based on the future worth method at rate of interest = 15%.

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